Dean Wu

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Education

- **University of Toronto** — Mississauga, Canada
  Bachelor of Science in Computer Science, Statistics and Mathematics; — *Sep. 2023 – May 2027*

Experience

- **UTASR (University of Toronto Autonomous Scale Racing)** — Toronto, Canada
  Software & Optimization Engineer — *Oct 2025 - Present*
  - **Nvidia Jetson Orin NX**: Architecting a stable, high-performance runtime on a headless Jetson Orin NX , using Python multiprocessing and IPC queues to manage parallel ML and Data Engineering workloads.
  - **Automation**: Developing a complete system automation and monitoring framework integrating Bash, cron, tegrastats, and htop to ensure low-overhead logging (<1ms) and rapid watchdog restarts (<5s).
  - **Remote System**: Implementing a robust remote orchestration system via SSH , delivering automated scripts for system startup, health checks, and repository updates.
- **Robotics Team | Loyola C.S.S.** — Mississauga, Canada
  Lead Software Engineer — *Sep. 2022 - Mar. 2023*
  - **Automated Doorway System**: Developed C++ firmware for an Arduino microcontroller to manage the control logic and actuation of an automated doorway system in a remote-controlled floor hockey robot.
  - **Physical Structure**: Designed and fabricated a custom chassis to house and integrate the control system (Arduino, sensors, actuators) for a remote-controlled floor hockey robot.
- **Game Making Team | Loyola C.S.S.** — Mississauga, Canada
  Lead Software Engineer/Project Manager — *Dec. 2022 - May 2023*
  - **Project Management**: Managed the project's development using Scrum methods. This involved planning work in focused sprints and leading regular team check-ins to meet all deadlines with a high-quality result
  - **Software Implementations**: Lead the game development by architecting the project in GameMaker Studio 2, coding core systems (such as movement, fighting, quests) in GML, and leading quality assurance in map design and collaborative team code reviews.

Projects

- **Discord AI**: Developed a multimodal Discord AI bot with autonomous agents for web search, stock/crypto tracking, weather forecasting, and gaming statistics using Python, LangChain, Vision AI, and Upstash Redis for persistent memory
- **Project Mura**: Developed a top-down 2D action game using GameMaker Studio 2 with custom GLSL shader programming.
- **deanwu.me**: Developed and deployed a responsive personal portfolio website using ReactJS and MUI(Material UI).

Programming Skills

- **Languages**: Python, Java, GameMaker Language, RISC-V Assembly, C++
- **Technologies**: React, MUI, NPM, Upstash, NodeJS, Raspberry Pi, Arduino, Git